

Position-Dependent Curvature for Hyperbolic Embeddings of Medical Ontologies

A Research Proposal

Abstract

Modern medical large language models represent disease concepts in flat Euclidean embedding spaces, despite the established result that tree-structured data cannot be embedded in Euclidean space without distortion that grows logarithmically with the size of the tree. Recent work has applied constant-curvature hyperbolic embeddings to medical ontologies (Beam et al., 2018; Yang et al., 2025; Kim et al., 2026), demonstrating substantial improvements over Euclidean baselines. However, these methods impose a single global curvature on heterogeneous ontologies whose local structural properties vary substantially across regions. We empirically verify this heterogeneity in ICD-10-CM: across the 255 sections of the ontology with at least three internal diagnosis nodes, mean branching factor varies from 2.33 to 10.00 — a 4.29-fold range — and subtree size per node varies tenfold across chapters. We propose to derive and evaluate position-dependent curvature κ for hyperbolic embeddings of medical ontologies, with κ derived from node-level structural statistics. We propose two complementary instantiations: a tree-anchored variant in which curvature at each embedded node is fixed by tree statistics computed offline, and an alternating-optimization variant in which curvature and embeddings are jointly refined through iterated updates. We compare both against four baselines spanning Euclidean, constant-curvature hyperbolic, learnable global curvature, and mixed-curvature product spaces. We address the chicken-and-egg problem between curvature and embeddings explicitly through the tree-anchored formulation, and we address tractability of distance computation through line-integral approximation of geodesic distance with empirical accuracy verification. Downstream evaluation targets rare-disease retrieval with strict subtree holdout to prevent leakage across the HPO, Mondo, and Orphanet resources. Phase 1 (data pipeline) is complete; the proposal outlines remaining phases with realistic timeline.

1. Introduction and Motivation

Rare-disease misdiagnosis is a documented clinical crisis. The average rare-disease patient sees roughly seven specialists over five or more years before receiving a correct diagnosis. A significant component of this failure mode is informational: clinicians rely on lookup tools and AI-augmented decision support that struggle to navigate the implicit hierarchy of disease classifications. When a clinician searches for a candidate diagnosis, current systems poorly surface adjacent diagnoses — diseases that branch off one or two steps away in the medical taxonomy — leading to systematic misses for rare conditions whose symptom patterns overlap with more common diseases.

Modern medical large language models inherit and amplify this limitation. These models represent disease concepts as dense vectors in flat Euclidean embedding spaces. However, medical ontologies — including ICD-10-CM, SNOMED-CT, and Mondo — are fundamentally hierarchical, and a classical theorem in metric embedding

establishes that trees cannot be embedded in Euclidean space without distortion that grows logarithmically in the size of the tree (Linial et al., 1995). Two diseases that are siblings in the medical taxonomy must therefore land either too close, causing confusion, or too far apart, failing to capture relatedness, when embedded in flat space.

Hyperbolic geometry, with its exponential volume growth, is the natural setting for tree-structured data. Sarkar (2011) showed that any tree admits a $(1+\epsilon)$ -distortion embedding into the Poincaré disk in linear time. Nickel and Kiela (2017) demonstrated that five-dimensional Poincaré embeddings substantially outperform two-hundred-dimensional Euclidean embeddings on hierarchy reconstruction. Beam et al. (2018) ported this technique to the ICD-9 hierarchy on a sixty-three-million-member insurance claims dataset, demonstrating that two-dimensional Poincaré embeddings perform comparably to one-hundred-dimensional Euclidean embeddings on the task of reconstructing chapter structure. Recent work (Yang et al., 2025; Kim et al., 2026) has continued to extend hyperbolic methods to medical contexts. Across this body of work, however, the curvature parameter is global — typically $K = -1$ — and treated as a hyperparameter rather than as a quantity derived from the structural properties of the data. Phylogenetic embedding work (Macaulay et al., 2023) found that improvement saturates around $K = -1$ across uniform-tree data, suggesting that global curvature tuning is not the main lever; for heterogeneous data, position-dependent curvature is the natural extension.

Medical ontologies are structurally heterogeneous at granularity finer than chapter level. The volumetric requirement at a node of branching factor b is exponential in b , implying that the optimal local curvature should depend on local structure. Forcing all regions of a heterogeneous ontology into a single global curvature is geometrically suboptimal, and the resulting distortion concentrates in regions where the ontology is least uniform — often the same regions where rare diseases reside. This proposal develops the case that medical ontology embeddings should use a position-dependent curvature κ tied to local structural statistics, with two distinct algorithmic instantiations.

2. Background

2.1 Hyperbolic Geometry and Tree Embeddings

Hyperbolic space is the unique simply connected complete Riemannian manifold of constant negative sectional curvature. In the Poincaré ball model, the manifold is the open unit ball in n -dimensional Euclidean space equipped with the conformally rescaled metric:

$$g_x = (4 / (1 - \|x\|^2)^2) \cdot g^E$$

The induced distance between two points u, v in the ball is:

$$d(u, v) = \operatorname{arcosh}(1 + 2\|u - v\|^2 / [(1 - \|u\|^2)(1 - \|v\|^2)])$$

Volume in n -dimensional hyperbolic space grows exponentially with radius — $V(r)$ scales as $\exp((n - 1)r)$ — in contrast to polynomial growth in Euclidean space. This volumetric exponential growth is what enables low-distortion tree embeddings. Sarkar's construction provides an explicit linear-time algorithm for embedding any weighted

tree into the Poincaré disk with worst-case distortion at most $1 + \varepsilon$ for appropriately scaled edges.

2.2 Hyperbolic Embedding Learning

Nickel and Kiela (2017) introduced learning Poincaré embeddings via Riemannian stochastic gradient descent. Given a set D of observed parent-child or related pairs, the loss is a softmax-style negative-sampling formulation:

$$L(\Theta) = \sum_{(u,v) \in D} \log [\exp(-d(u, v)) / \sum_{v' \in N(u)} \exp(-d(u, v'))]$$

The Riemannian gradient update is:

$$\theta_{t+1} \leftarrow \text{proj}(\theta_t - \eta_t \cdot ((1 - \|\theta_t\|^2)^2 / 4) \cdot \nabla^E L)$$

with the projection operator ensuring embeddings remain inside the open unit ball.

3. Empirical Motivation: Heterogeneity in ICD-10-CM

We parsed the FY2025 ICD-10-CM Code Tables tabular XML release from CMS into a unified tree with one synthetic root, 22 chapters, 296 sections, and 46,498 diagnosis nodes for a total of 46,817 nodes and 46,816 edges. The often-quoted figure of 70,000+ ICD-10 codes refers to billable codes including encounter-modifier variants; for the hierarchical embedding task the relevant count is the 46,500 distinct clinical concepts.

Across the 10,775 internal nodes, branching factor has mean 4.34, median 3, and maximum 34. Aggregated to chapter level, mean branching factor varies modestly: from 3.67 to 5.32, a 1.45-fold range. Within-chapter standard deviations lie between 1.3 and 2.8, indicating each chapter is internally fairly uniform.

Aggregated to section level, heterogeneity is substantially larger. Across the 255 sections with at least three internal diagnosis nodes, mean branching factor ranges from 2.33 to 10.00, a 4.29-fold range with median 4.25 and standard deviation of section means equal to 1.10. Chapter-level smoothing masks real fine-grained structural variation; section-level statistics expose it.

We computed a complementary measure: total descendant count (subtree size) for every internal node, summarized by chapter. Maximum subtree size by chapter ranges from 37 (Chapter 1) to 361 (Chapter 19) — approximately a 10-fold range. Some chapters contain densely-packed sub-branches with hundreds of descendants beneath a single node; others have shallower, sparser structure. This depth-density variation is complementary to branching-density variation, capturing a distinct geometric property.

Together, these analyses establish that ICD-10-CM exhibits real, multi-dimensional heterogeneity: branching heterogeneity at section granularity ($\sim 4\times$ ratio) and subtree-density heterogeneity at chapter granularity ($\sim 10\times$ ratio). Both correspond to local geometric requirements that constant-curvature embeddings cannot simultaneously satisfy across all regions.

4. Problem Formulation

Let $T = (V, E)$ be a medical ontology with rooted tree structure. For each internal node $v \in V$, define $b(v)$ (branching factor) and $s(v)$ (subtree size). We seek an embedding $f: V \rightarrow M$ into a Riemannian manifold (M, g) such that the graph distance d_T is approximated by the Riemannian distance d_g with low multiplicative distortion, the manifold is a Cartan-Hadamard manifold so geodesics are unique and exponential maps are global diffeomorphisms, sectional curvature κ varies as a function of node-level statistics (b, s) rather than being constant, and the resulting Riemannian optimization remains computationally tractable.

We explicitly anchor curvature to node-level tree statistics rather than to embedding-region density. This is a deliberate methodological choice that resolves the circular dependency that would otherwise arise: if κ at position x depended on the density of embedded points near x , and embedding positions depended on κ through gradient updates, the system would be ill-defined without an alternating scheme. Anchoring κ to tree-derived statistics decouples the dependency: tree statistics are computed once, offline, before any embedding work begins.

5. Proposed Methods

5.1 Mathematical Framework

We propose a conformally Euclidean variable-curvature manifold:

$$g_x = (1 / \lambda(x)^2) g^E$$

where $\lambda(x)$ is a smooth positive function generalizing the conformal factor $(1 - ||x||^2)/2$ of the standard Poincaré ball. The induced sectional curvature depends on λ via a quasi-linear elliptic operator. To guarantee non-positive curvature globally and to preserve completeness, we restrict $\lambda(x)$ to a parametric family for which these conditions can be verified analytically.

We derive the parametric form for κ from a volume-matching argument. For a node of branching factor b at depth d , the tree contains approximately b^d leaves below it. Hyperbolic volume at radius r in n dimensions grows as $\exp((n - 1)r\sqrt{|K|})$. Identifying tree depth with embedding radius up to a scale factor, matching volume to leaf count gives the leading-order relationship $|K| \propto (\log b)^2$. We complement this branching-derived term with a subtree-size correction reflecting depth-density:

$$\rho(v) = \alpha \cdot b(v) + \beta \cdot \log(s(v))$$

$$\kappa(v) = \kappa_0 \cdot (1 + \gamma \cdot \rho(v))$$

with $\kappa_0 < 0$ a baseline curvature and α, β, γ learned scalar parameters with constraints ensuring $\kappa(v) < 0$ for all v in the relevant ranges. Full derivation, including verification of Cartan-Hadamard preservation for the parametric form, is a planned deliverable.

5.2 Two Algorithmic Instantiations

We propose two complementary methods to evaluate position-dependent curvature:

Tree-anchored curvature. For each node v , compute $\rho(v)$ once from tree statistics and fix $\kappa(v) = \kappa_0(1 + \gamma\rho(v))$ for the duration of training. The Riemannian gradient update uses the node-specific curvature value at each embedding position. This formulation is fully decoupled from embedding density and avoids any circular dependency. It is also computationally efficient: no curvature recomputation during training is required.

Alternating optimization. Initialize embeddings using tree-anchored curvature. After a fixed number of training epochs, recompute a smoothed curvature field $\tilde{\kappa}(x)$ from kernel density of current embedding positions, then continue training using $\tilde{\kappa}(x)$ in place of the node-level values. Alternate between embedding updates and curvature updates until convergence. This formulation allows curvature to refine based on emergent geometric properties of the embedded ontology while remaining anchored to tree-derived statistics through the smoothing prior.

5.3 Tractable Distance Computation

For a general conformally Euclidean variable-curvature manifold, geodesic distance does not admit a closed form. Numerical ODE integration for every pairwise distance during training is prohibitive at ontology scale ($\sim 50,000$ nodes, millions of pairs per epoch). We address this with line-integral approximation: the distance between u and v is computed as the integral of the local metric along the straight Euclidean segment connecting u and v , rather than along the true geodesic. For small to moderate displacements in regions of slowly varying curvature, this approximation is close to the true geodesic distance. We empirically characterize the approximation error against numerical ODE integration on a held-out subset of pairs before committing to the approximation in the main training loop.

6. Experimental Plan

6.1 Phase 1: Data Pipeline (completed)

Acquired and parsed FY2025 ICD-10-CM tabular XML from CMS into a unified tree (46,817 nodes, 46,816 edges). Computed branching and subtree-size statistics at multiple granularities, establishing the empirical heterogeneity reported in Section 3. Generated diagnostic visualizations and persisted the parsed tree along with per-node feature statistics for use in subsequent phases.

6.2 Phase 2: Baselines

Four baseline methods, each trained and evaluated under identical protocol:

(i) **Euclidean embedding** with cosine similarity, justifying the need for hyperbolic geometry at all. (ii) **Constant-curvature Poincaré ball** with $K = -1$, the standard baseline introduced by Nickel and Kiela. (iii) **Learnable global curvature**, where K is a single trainable scalar parameter, addressing the question of whether tuning a single global value can recover the benefits of variable curvature. (iv) **Mixed-curvature product spaces** (Gu et al., 2019), which address heterogeneity by combining multiple constant-curvature factor spaces. We omit hyperbolic entailment cones (Ganea et al., 2018) from main experiments, as they address asymmetric hierarchical relations — an

orthogonal improvement axis from heterogeneity of branching density. They could be combined with our variable-curvature manifold as future work.

6.3 Phase 3: Novel Methods

We implement both the tree-anchored and alternating-optimization formulations described in Section 5.2. Each is trained with the same negative-sampling loss and evaluation protocol as the baselines, controlling for embedding dimension and learning rate. Per-chapter and per-section breakdowns are reported to localize where each method helps and where it does not. An ablation varying γ (curvature variation amplitude) across $\{0, 0.25, 0.5, 1.0, 2.0\}$ characterizes sensitivity to the parametric form.

6.4 Phase 4: Downstream Evaluation

Rare-disease retrieval benchmark using HPO phenotype annotations, Mondo Disease Ontology alignments, and Orphanet rare-disease entities. We design the evaluation explicitly to prevent leakage: a held-out test set is constructed by selecting entire Orphanet sub-branches such that the test diseases and their HPO annotations are excluded from training data. The model is trained on the remaining ontology fragments and must retrieve held-out diseases from their characteristic symptom sets at evaluation time. Metrics include top-k recall and mean reciprocal rank, reported overall and stratified by disease rarity, branching density of the local ontology region, and tree depth.

6.5 Datasets and Compute

Open and immediately accessible: ICD-10-CM (CMS, public domain), Mondo Disease Ontology (CC-BY 4.0), HPO with HPOA annotations, Orphanet, MedQA, Med-HALT. Free with registration: UMLS Metathesaurus. Credentialed: MIMIC-IV via PhysioNet, used as supplementary validation. All training is feasible on a single A100 GPU. Phase 2 baselines fit on a T4-class GPU for dimensions up to 100; Phase 3 variable-curvature training requires A100 for dimension 50, well within the relevant range.

7. Expected Contributions

(C1) The first empirical characterization of structural heterogeneity in ICD-10-CM at section-level granularity. (C2) An explicit derivation of position-dependent curvature κ for hierarchical embeddings of medical ontologies, motivated by node-level branching factor and subtree-size statistics, with verified Cartan-Hadamard preservation. (C3) An implemented and benchmarked variable-curvature Riemannian optimization framework, including line-integral distance approximation and empirical accuracy characterization. (C4) Comparative evaluation of two algorithmic instantiations (tree-anchored versus alternating-optimization) against four baselines on ICD-10-CM reconstruction and rare-disease retrieval with strict subtree holdout. (C5) Open-source release of code, trained embeddings, and parsed ontology data.

8. Risks and Mitigations

Cartan-Hadamard preservation. The parametric family for $\lambda(x)$ must be chosen so that the induced curvature is non-positive everywhere. We restrict to a family with analytic guarantees rather than relying on per-batch verification. **Distance computation cost.** Numerical geodesic integration is prohibitive; we use line-integral approximation with empirical accuracy verification on a held-out pair sample. **Alternating-optimization convergence.** The alternating scheme is not guaranteed to converge in general. We monitor embedding stability across iterations and report whether convergence is observed empirically; if not, we report performance at fixed iteration counts and discuss the implications. **Empirical generalization.** Improvements may not be uniform across chapters or dimensions. We report per-chapter and per-section results to localize benefits.

9. Timeline

Phase 1 (data pipeline): completed. Phase 2 (baselines): weeks 1-6. Phase 3 (novel methods): weeks 7-17. Math derivations in parallel: weeks 8-18. Phase 4 (rare-disease retrieval): weeks 16-22. Writing and submission: weeks 22-26. Target: Regeneron Science Talent Search submission deadline (mid-November 2026).

10. Conclusion

Hyperbolic embeddings are the appropriate geometry for hierarchical medical data, but existing methods uniformly assume constant curvature, ignoring substantial heterogeneity in real medical ontologies — heterogeneity we have empirically verified in ICD-10-CM. We propose two complementary instantiations of position-dependent curvature, both anchored to tree statistics to avoid the circular dependency between curvature and embeddings. We compare against a comprehensive set of baselines including Euclidean, constant-curvature Poincaré, learnable global curvature, and mixed-curvature product spaces. We address tractability via line-integral distance approximation, and design downstream evaluation with strict subtree holdout to prevent leakage. The empirical heterogeneity established in this proposal grounds the project in observed structural properties of the data; the methodological design explicitly addresses the principal concerns raised in prior review.

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